

- 7 Read the passage below and answer the questions that follow.

Interaction of X-ray with matter

X-rays have wavelengths of 10^{-12} to 10^{-9} m. It can undergo different interaction with matter. Depending on the energy of X-ray and the material of the target metal, *photoelectric effect*, *Compton scattering* or *pair production* may occur. Fig. 7.1 shows the dominant type of interaction depending on the atomic number of the target and the energy of X-ray.

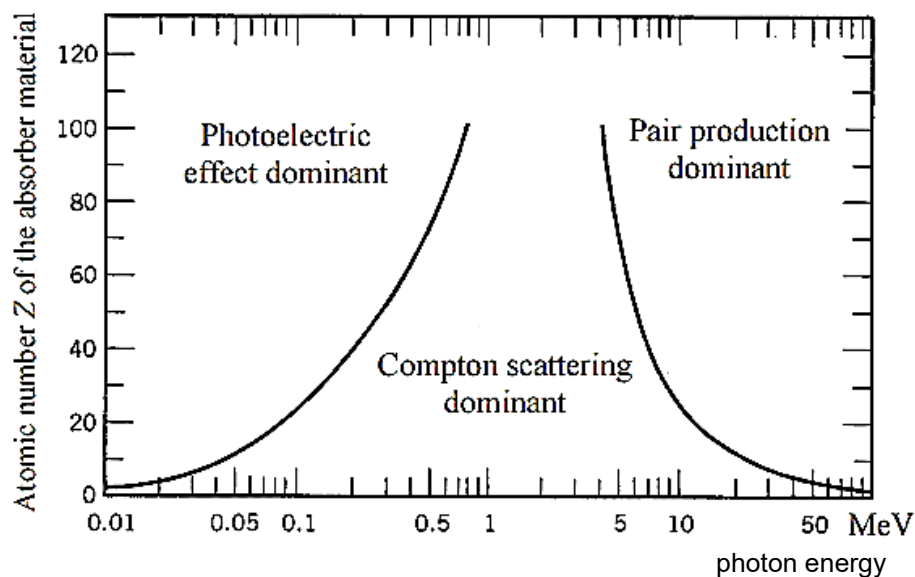


Fig. 7.1

- (a) Electromagnetic radiation of wavelength of 6.2×10^{-12} m is incident on an iron-54 ($^{54}_{26}\text{Fe}$) target.

- (i) Determine the energy of the photon.

energy = MeV [3]

- (ii) Hence, state the dominant form of interaction between the X-ray photon and the target.

..... [1]

[Turn over

- (b) During Compton scattering, an incident X-ray photon interacts with an outer loosely bound electron, and the electron recoils. The scattered X-ray photon has a lower photon energy than the incident photon.

In a Compton scattering experiment, an X-ray beam is incident on graphite. Fig. 7.2 shows an incident photon of wavelength λ_0 colliding with a stationary electron and the scattered photon of wavelength λ moving at scattering angle θ . The electron recoils at an angle ϕ .

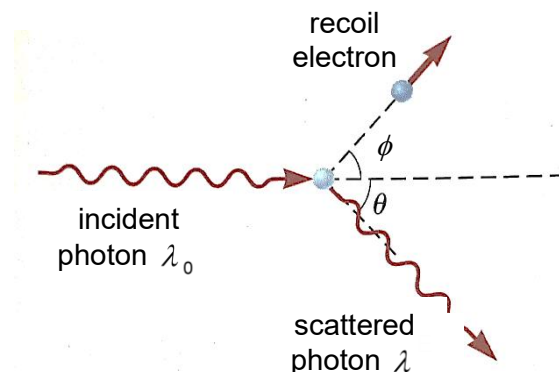


Fig. 7.2

The relationship between E , the energy of the scattered photon and E_0 , the energy of the incident photon, is given by

$$\frac{1}{E} - \frac{1}{E_0} = \frac{1}{m_e c^2} (1 - \cos \theta)$$

where m_e is the mass of the electron, and
 c is the speed of light.

The Compton shift is the difference in wavelength between the incident photon and the scattered photon.

- (i) Show that the Compton shift $\lambda - \lambda_0$ is given by the expression

$$\lambda - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$$

where λ_0 is the wavelength of the incident photon,

λ is the wavelength of the scattered photon, and

[1]

- (ii) State the angle $\theta_{\max}$ for the greatest Compton shift.

$$\theta_{\max} = \dots\dots\dots^\circ \quad [1]$$

- (iii) Hence or otherwise, sketch on Fig 7.3 the variation with θ of the Compton shift $\lambda - \lambda_0$ as θ varies from 0° to 180° .

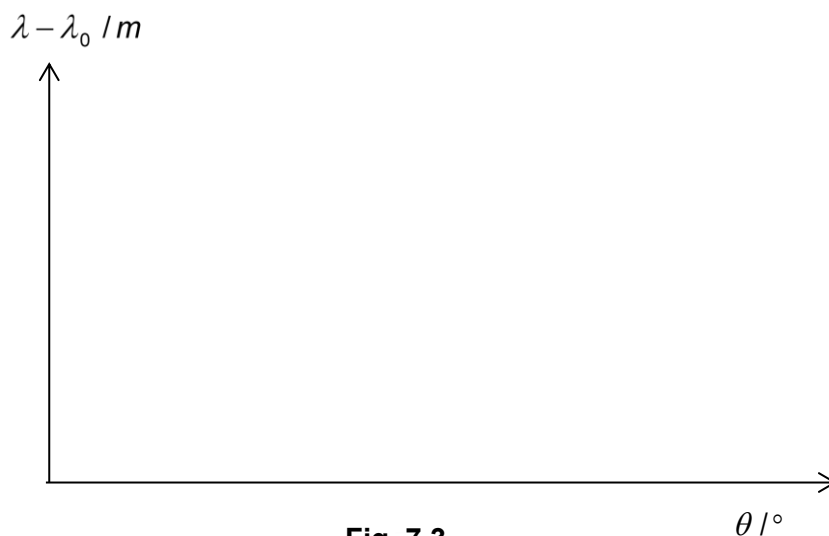


Fig. 7.3

[1]

[Turn over

- (iv) An X-ray of wavelength $7.09 \times 10^{-11} \text{ m}$ is incident on a graphite. A detector, placed at different scattering angle θ , recorded two intensity peaks except at $\theta = 0^\circ$. The first peak is at wavelength λ_0 while the second peak is at λ . Fig. 7.4 (a) to (d) shows the intensity peaks at various θ .

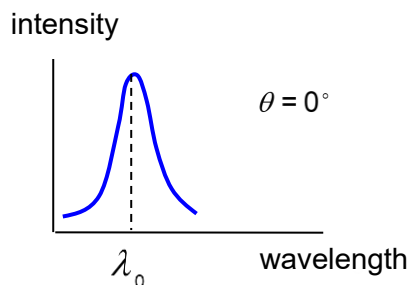


Fig. 7.4 (a)

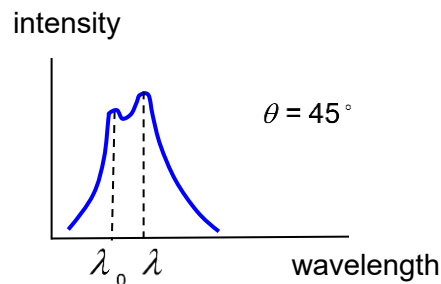


Fig. 7.4 (b)

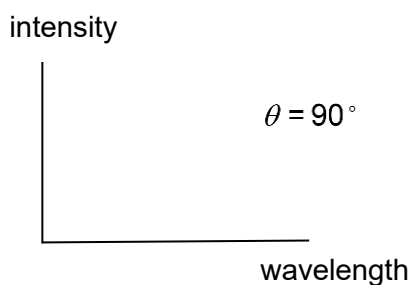


Fig. 7.4 (c)

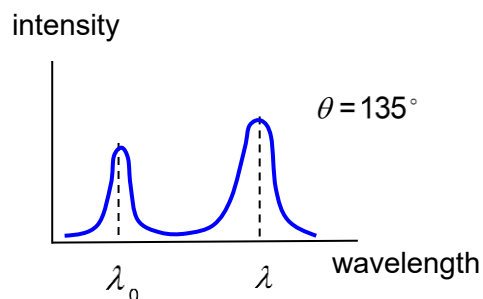


Fig. 7.4 (d)

1. Explain why there is only one peak when $\theta = 0^\circ$.

.....
 [1]

2. Deduce and sketch the line for $\theta = 90^\circ$ on Fig. 7.4 (c). [1]

- (v) The measurements taken in (c)(iv) at varying scattering angles θ when an X-ray of wavelength $7.09 \times 10^{-11} \text{ m}$ is incident on a graphite is as shown in Fig. 7.5.

$\theta / ^\circ$	45	70	95	110	135	160
$\lambda - \lambda_0 / 10^{-11} \text{ m}$	0.08	0.16		0.34	0.41	0.46
$1 - \cos \theta$	0.29	0.66		1.34	1.71	1.98

Fig. 7.5

- Complete the Fig 7.5 for $\theta = 95^\circ$. [1]

- (vi) Fig. 7.6 shows the variation of $1 - \cos \theta$ with $\lambda - \lambda_0$ based on the measurements in (c)(v).

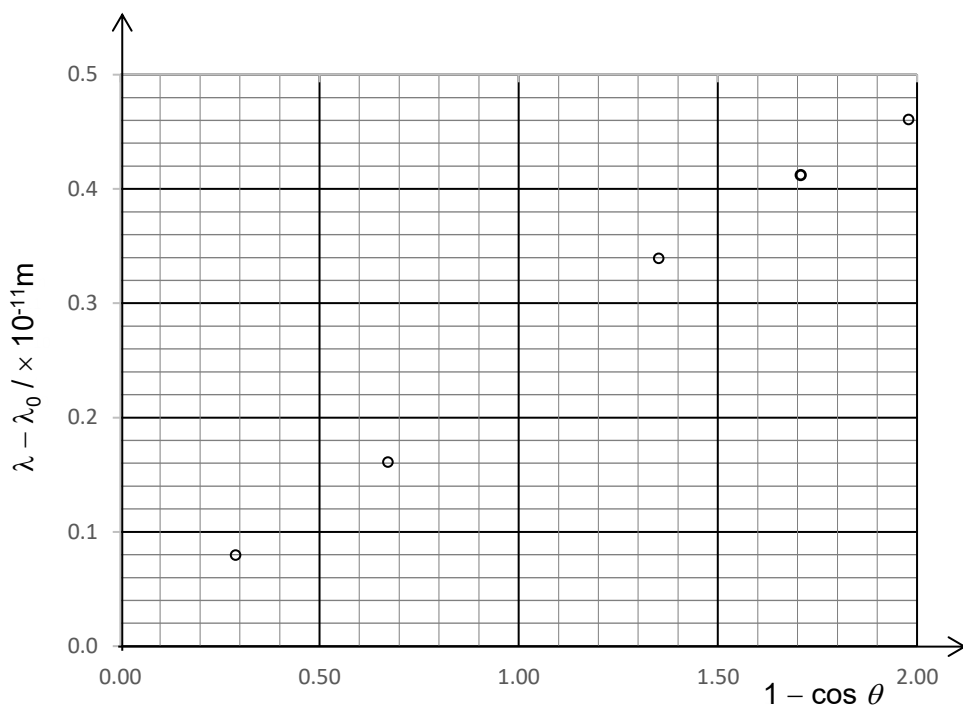


Fig. 7.6

On Fig. 7.6,

1. plot the point corresponding to $\theta = 95^\circ$, [1]
2. draw the line for best fit for all the points. [1]
3. Using Fig 7.6, determine the Compton wavelength of electron $\frac{h}{m_e c}$.

$$\frac{h}{m_e c} = \dots\dots\dots \text{ m [2]}$$

[Turn over

4. Use Fig. 7.6, determine the wavelength λ of the scattered photon at $\theta = 90^\circ$.

wavelength $\lambda = \dots\dots\dots$ m [2]

- (c) Pair production occurs when X-ray photon gets very close to the nucleus of an atom and experiences its strong field effects. The photon will annihilate or disappear, and subsequently produces a pair of an electron and positron. The probability of pair production increases with higher photon energy and higher atomic number.

In pair production, an X-ray photon annihilates to produce an electron and a positron. Both the electron and positron have the same rest mass.

- (i) Show that the minimum amount of energy needed to produce the electron and the positron is 1.64×10^{-13} J.

[2]

- (ii) Suggest why the photon that can cause pair production is likely to have an energy higher than 1.64×10^{-13} J.

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.....[1]